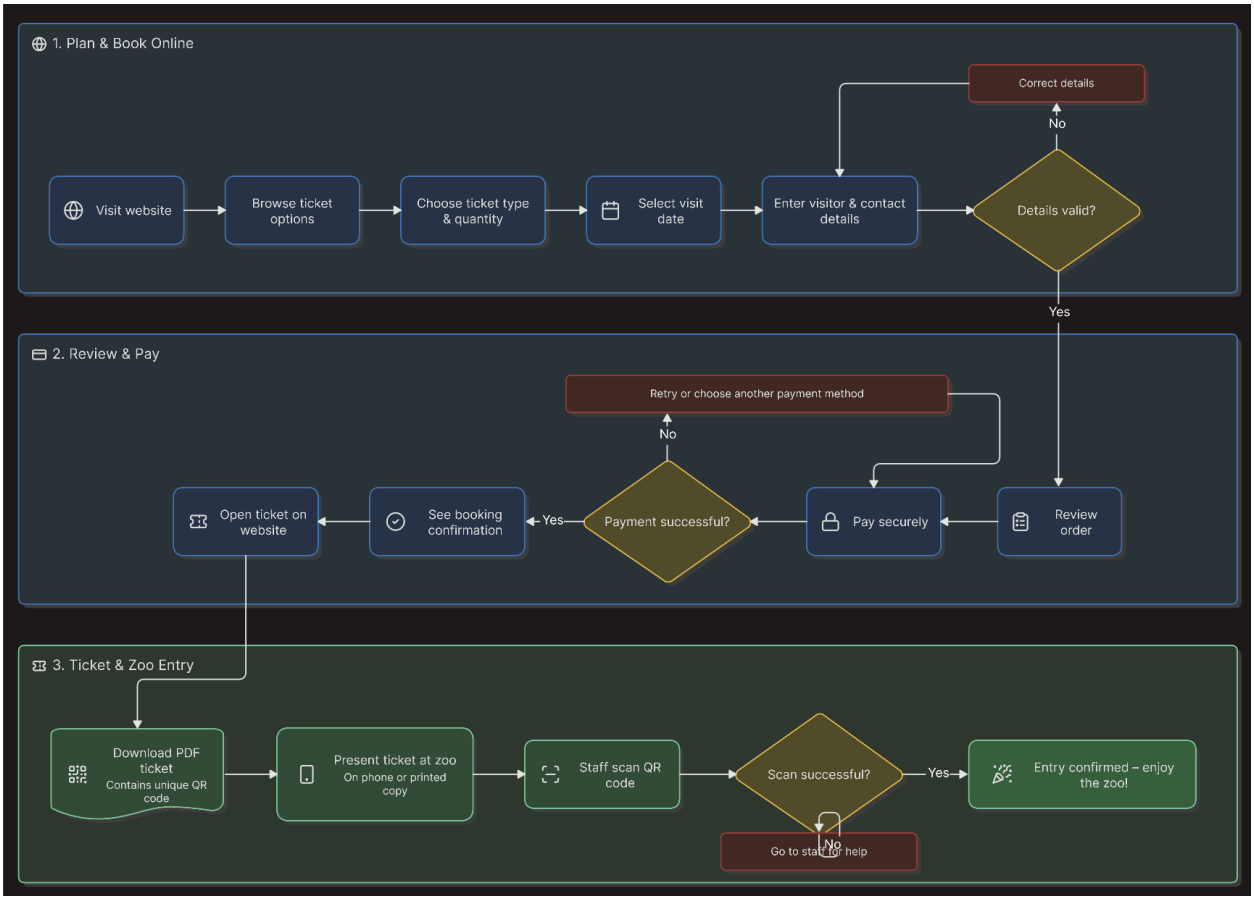
Digital Zoo ticket system

Digital Zoo Ticketing: UX & Service Design Case Study



Project Overview

- **Product:** Online Zoo Ticketing & Gate Entry System
- **Scope:** End-to-end User Flow, E-Commerce Checkout, and Physical Gate Integration
- **Primary Goal:** Reduce digital cart abandonment during checkout and eliminate physical bottlenecks at the zoo entrance through a unified, self-correcting booking and entry flow.

1. The Challenge

Modernizing a zoo's ticketing system requires bridging the gap between a digital e-commerce transaction and a physical on-site operational workflow. Visitors (often busy parents or tourists booking on mobile devices) face three primary friction points in legacy ticketing systems:

- **Complex Multi-Step Forms:** High drop-off rates when users encounter rigid form validation errors without losing previously entered data.
- **Payment Interruptions:** Failed transactions that force users to restart their entire ticket selection process from scratch.
- **Gate Bottlenecks:** Slow physical entry caused by lost confirmation emails, poor cellular reception at the entrance, or unreadable barcodes.

2. The Solution: A Three-Phase Service Architecture

To create a resilient, frictionless experience, the user journey was restructured into three distinct, modular phases—**1. Plan & Book Online**, **2. Review & Pay**, and **3. Ticket & Zoo Entry**—each equipped with dedicated error-recovery loops.

Phase	Core User Actions	System Validation & Recovery Mechanism	UX & Operational Value
1. Plan & Book Online	Visit website Browse options Select ticket type/quantity Pick visit date Enter contact details	**Details Valid?** If *No*, triggers an inline "Correct details" loop directly back to the contact form.	Preserves ticket selections and dates; prevents form-reset frustration.
2. Review & Pay	Review order Pay securely View booking confirmation Open ticket directly on website	**Payment Successful?** If *No*, routes to "Retry or choose another payment method" without leaving checkout.	Reduces cart abandonment by isolating payment failures from order review.
3. Ticket & Zoo Entry	Download PDF ticket (with unique QR code) Present on phone or print Staff scan QR Entry confirmed	**Scan Successful?** If *No*, directs visitor immediately to a dedicated "Go to staff for help" resolution path.	Supports offline access via PDF download; keeps main turnstiles moving.

3. Deep Dive: Key Architectural Decisions

Linear Progression with Micro-Validation (Phase 1)
Rather than overwhelming users with a single massive checkout page, the planning phase breaks decision-making into digestible sequential chunks: browsing options, specifying quantities, picking a calendar date, and inputting contact information. By placing the **"Details valid?"** decision gateway immediately after contact entry, the system catches formatting errors (e.g., invalid email addresses or missing phone numbers) before initializing the payment gateway.

Resilient Checkout & Instant Access (Phase 2)
In e-commerce, payment gateway timeouts or mistyped card details are leading causes of lost revenue. By designing an isolated retry loop (**"Retry or choose another payment method"**) that feeds directly back into the **"Pay securely"** action, users never lose their reserved date or ticket allocation. Furthermore, once payment succeeds, users do not have to hunt through spam folders for an email; the flow moves directly from **"See booking confirmation"** to **"Open ticket on website"**.

Bridging Digital to Physical Operations (Phase 3)
Zoo entrances often suffer from congested cellular networks. Prompting the user to **"Download PDF ticket (Contains unique QR code)"** immediately from the web confirmation screen ensures the ticket lives locally on their device (and allows traditional users to print a physical copy). At the gate, the physical workflow is designed for speed: staff scan the QR code for instant entry. If a screen is cracked or a ticket has already been redeemed (**"Scan successful? $\rightarrow$ No"**), the flow explicitly routes the guest to an exception-handling step (**"Go to staff for help"**) so the main entry queue remains unobstructed.

4. Key Takeaways & Impact Metrics

When presenting or evaluating this architecture in a portfolio or stakeholder review, this system design directly supports four core performance indicators:

- **Higher Checkout Conversion Rate:** Isolated error loops in Phases 1 and 2 prevent session drop-offs.
- **Zero Email-Dependency Delays:** Immediate on-site ticket rendering and PDF generation eliminate "missing confirmation email" support tickets.

- **Faster Gate Throughput:** Standardized QR scanning and offline-ready PDFs reduce average turnstile transaction time per family.
- **Clear Exception Handling:** Dedicated fallback paths for both digital failures (payment) and physical failures (gate scanning) ensure edge cases are handled gracefully.